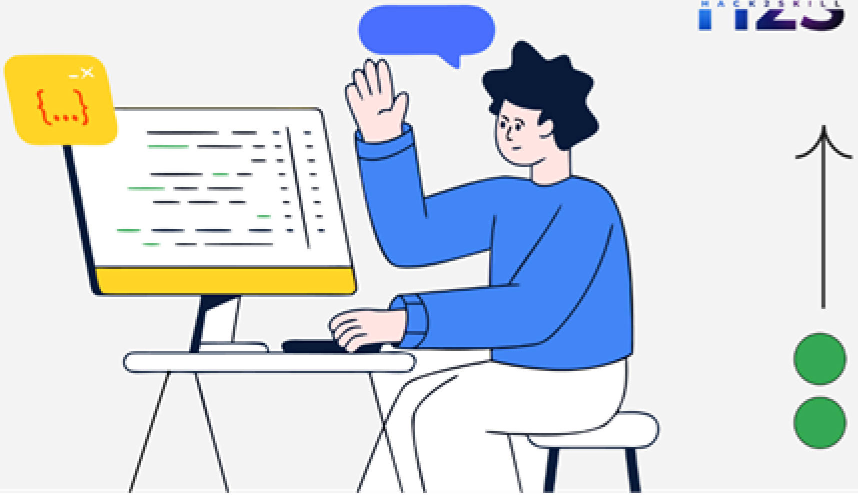
Build with AI

Solution Challenge



Team Details

a. Team name: **VISION ENCODERS**
b. Team leader name: **SRIMATHI B**
c. Problem Statement: **[Smart Resource Allocation] Data-Driven Volunteer Coordination for Social Impact**

Brief about our solution:

"QuickConnect" is a **centralized platform** that transforms scattered community data from reports, surveys, WhatsApp messages, and spreadsheets into **clear**, **prioritised insights** on the **most urgent needs** based on severity, location, and impact.

It uses a **smart matching system** that automatically connects the right volunteers to the right tasks based on their skills, availability, and location ensuring faster and more efficient response.

The system also provides a **real-time dashboard** for monitoring, updates, and manual overrides, giving NGOs full control while reducing coordination effort.

a. How different is it from any of the other existing ideas?

Most existing NGO tools:

1. Just store data (Google forms, Excel, CRMs)
2. Just list volunteers
3. Coordination is **manual + slow**

QuickConnect is different because:
-> It integrates multiple messy sources (Surveys, WhatsApp, reports, etc.)
-> It uses Gemini AI to convert messy inputs into structured, prioritized data.
-> It automatically assigns top 3 volunteers based on scoring (Availability, Distance & Skills)

Existing solutions manage data and volunteers separately; QuickConnect connects both intelligently.

b. How will it be able to solve the problem?

Current Problem:

Scattered data -> No clarity
Manual Coordination -> Delays
Wrong Volunteers -> Inefficient response

Our solution fixes the problem with:

Centralization -> All data in one place
Prioritization Engine -> Identifies most urgent needs
Smart Matching -> Assigns best fit volunteers instantly
Dashboard + Overrides -> Ensures control and flexibility

We reduce decision time and automate coordination, turning chaos into actionable response.

USP of the proposed solution

1. **Intelligent Matching Engine**
   Skill + Location + Availability based assignment

2. **Multi-Source Data Integration**
   Handles unstructured real-world data (paper, WhatsApp, etc.) with the help of AI.

3. **Hybrid Control System**
   Automatic assignment + Admin Override

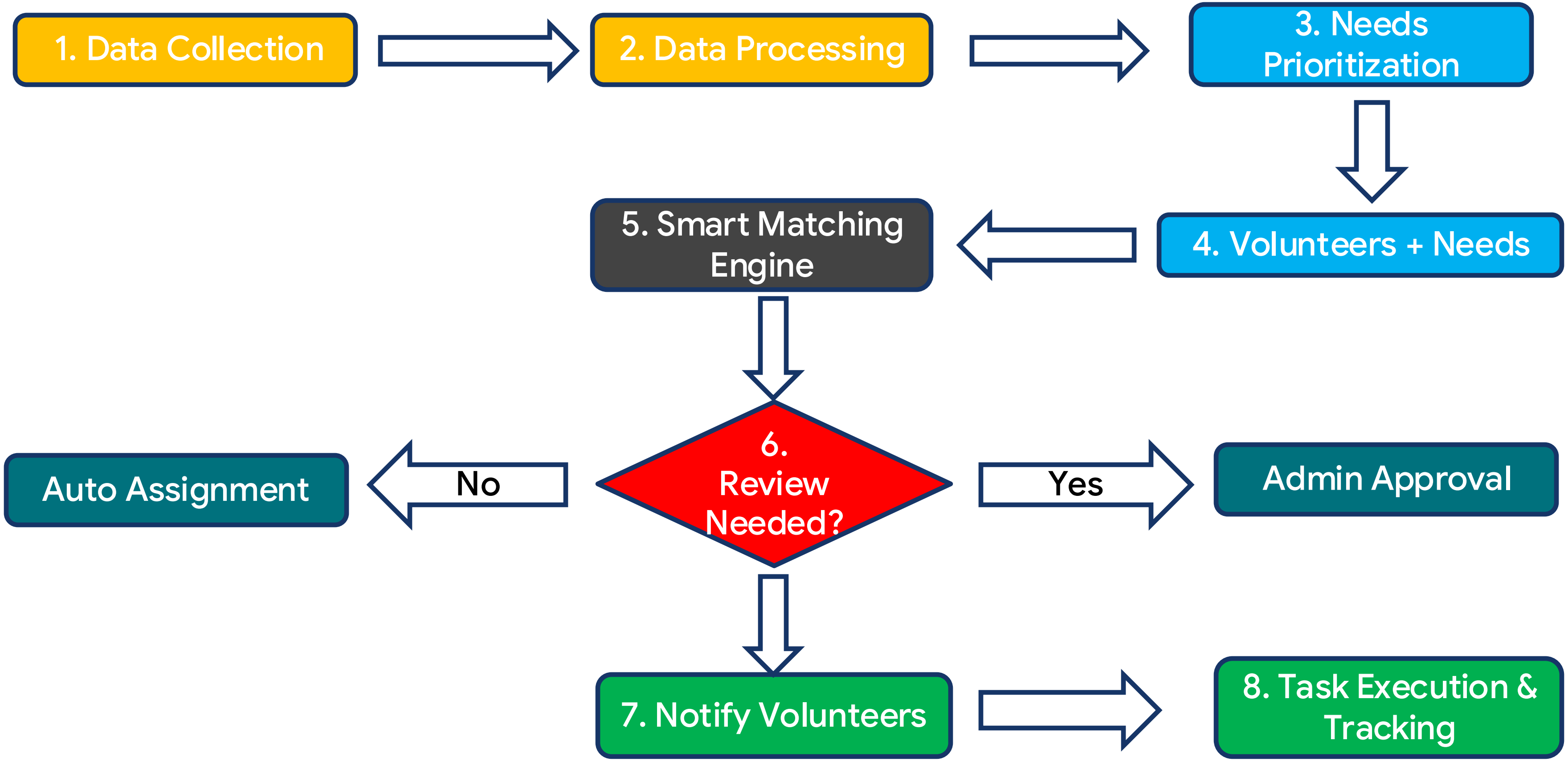
4. **Notification System + Feedback Loop**
   Real-Time notification and live updates via feedback

5. **Live Visualization**
   Visualize needs and volunteer assignments, enabling location-aware coordination

List of features offered by the solution

1. Multi-Source Data Integration
2. Centralized Data Management
3. AI-based Intelligent Need Prioritization
4. Rule-based Smart Volunteer Matching Engine
5. Real-Time Dashboard
6. Admin Control and Override
7. Volunteer Interface
8. Real-Time Notification System
9. Task Tracking and Status Updates
10. Geo-Based Optimization

Process flow diagram



Wireframes/Mock diagrams of the proposed solution

Quick Connect

🚨 Needs

Flood Relief Food — high — Match
📍 Velachery

Medical Help — medium — Match
📍 Tambaram

👥 Volunteers

Ravi	available
Priya	available
Arun	available
Kavin	available

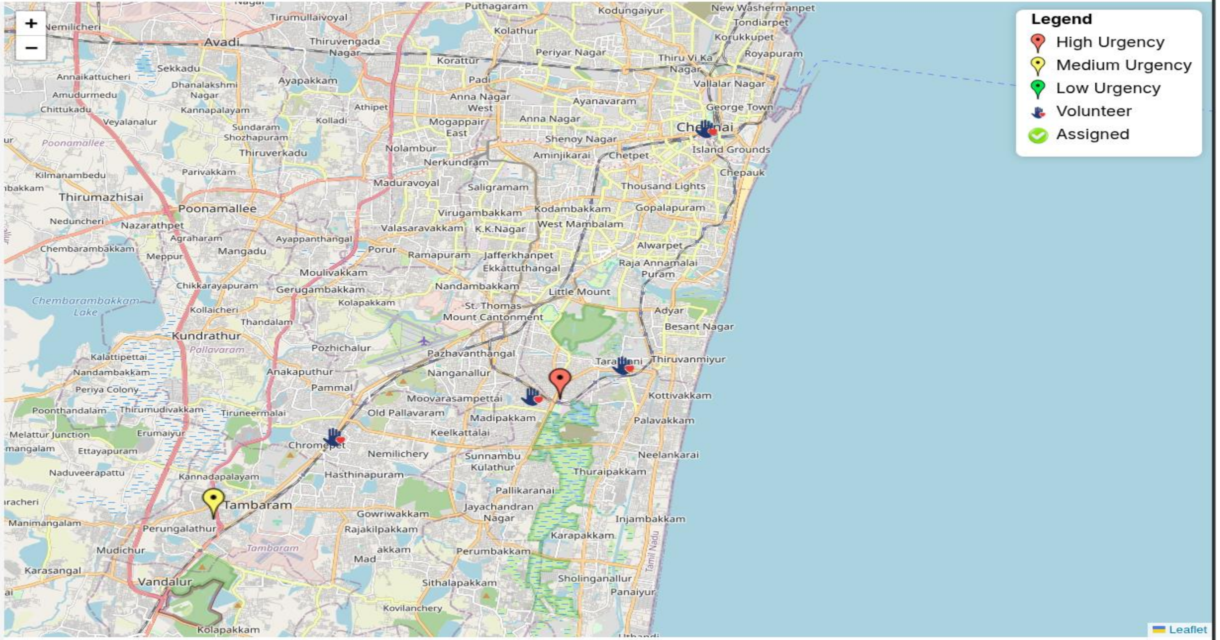
📌 Assignments

No assignments yet

⚙️ Demo Purpose

Clear All Assignments

Add Need



Multiple needs with **urgency** and **unassigned volunteers** before matching

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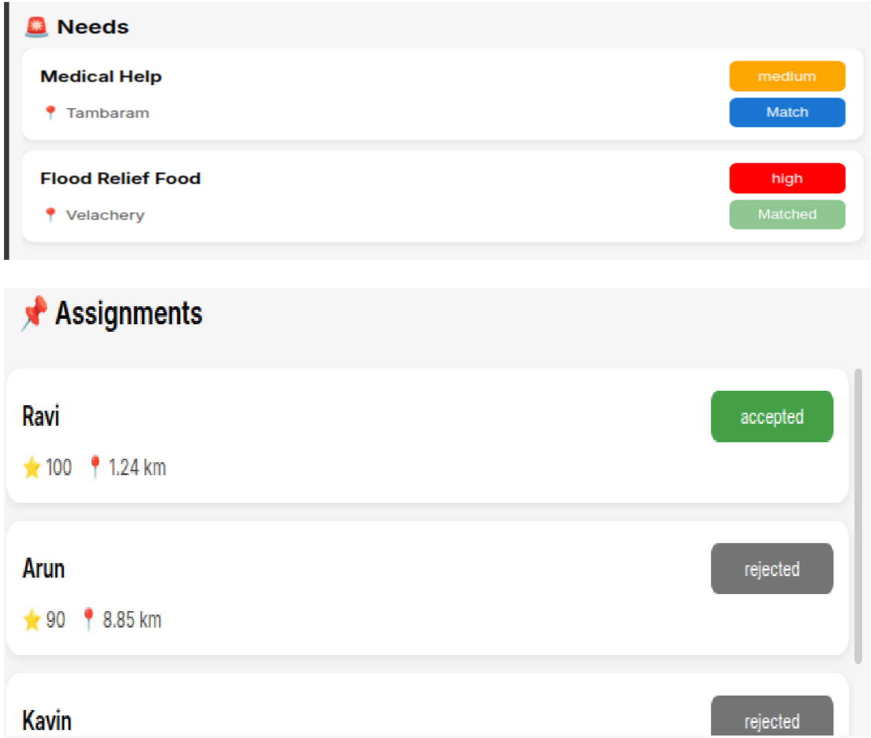
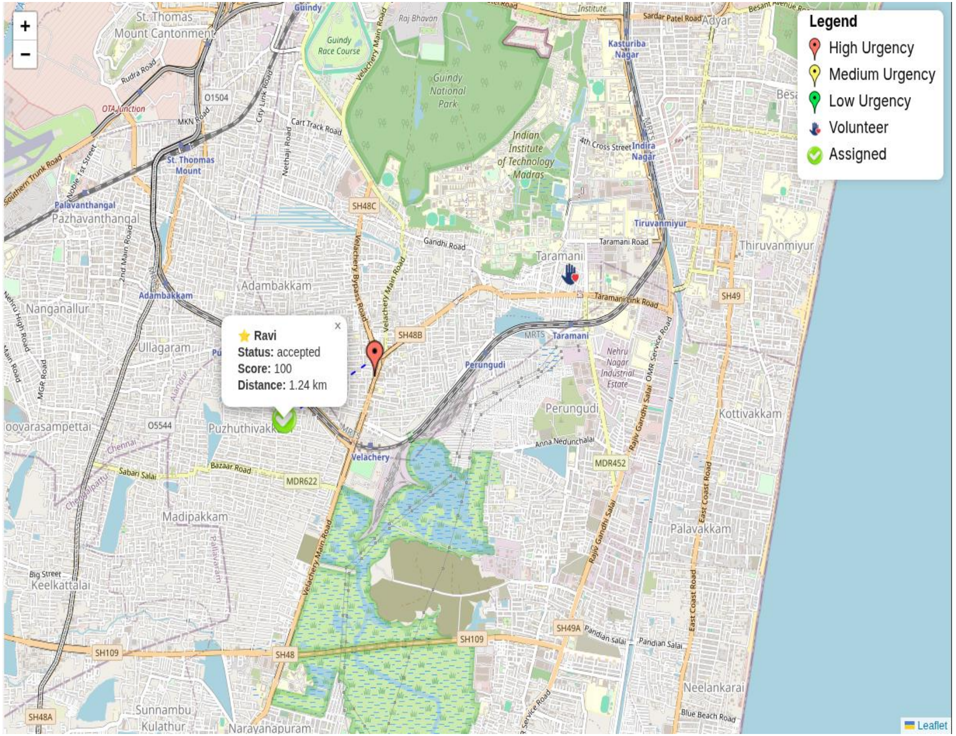
H2S

Wireframes/Mock diagrams of the proposed solution

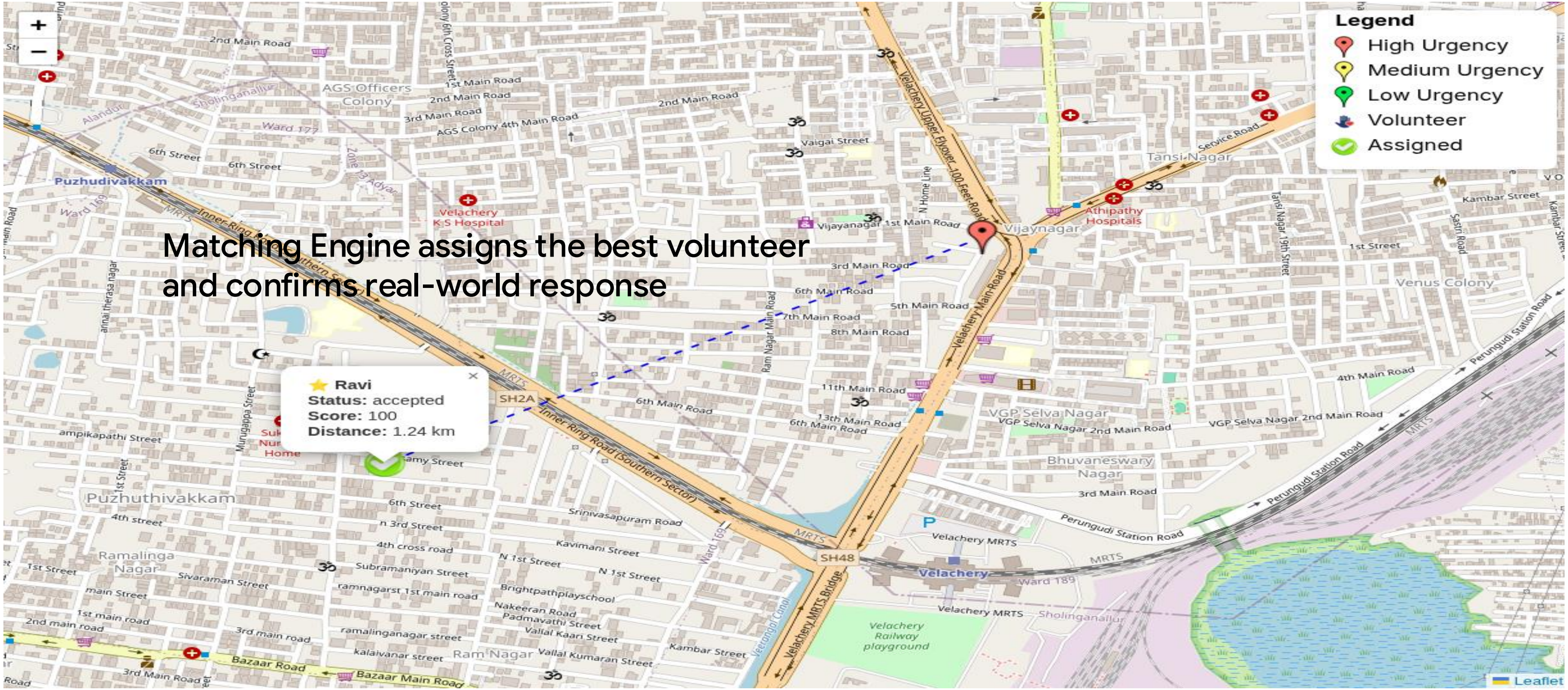




Smart matching engine ranks the top 3 volunteers based on availability, skills, and distance.

Live visualization of task assignment and its details

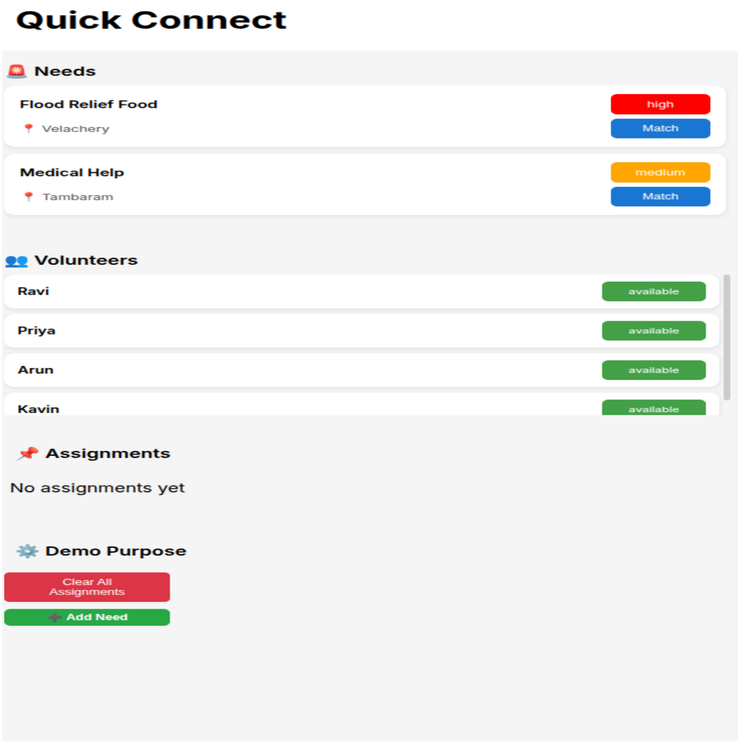
Wireframes/Mock diagrams of the proposed solution



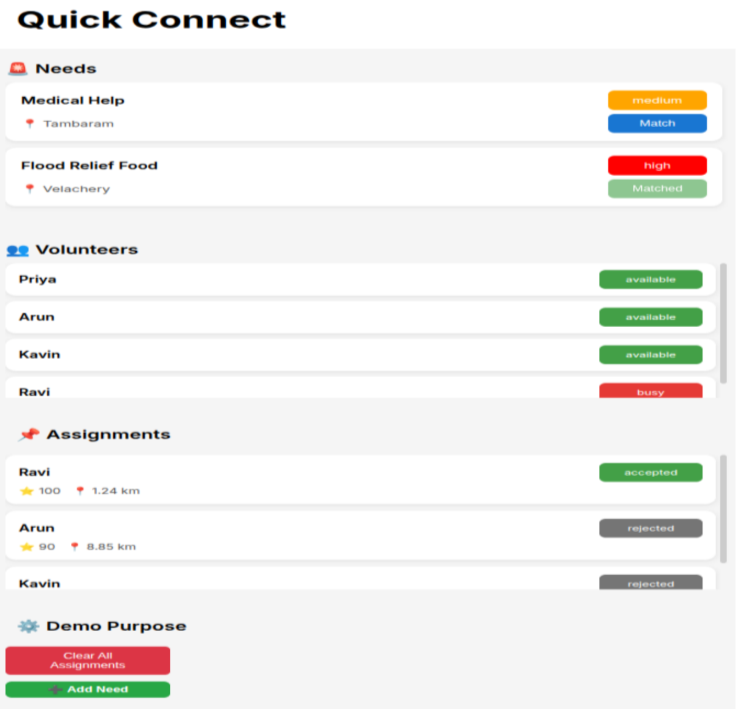
From request -> to verified response in real-time

Wireframes/Mock diagrams of the proposed solution

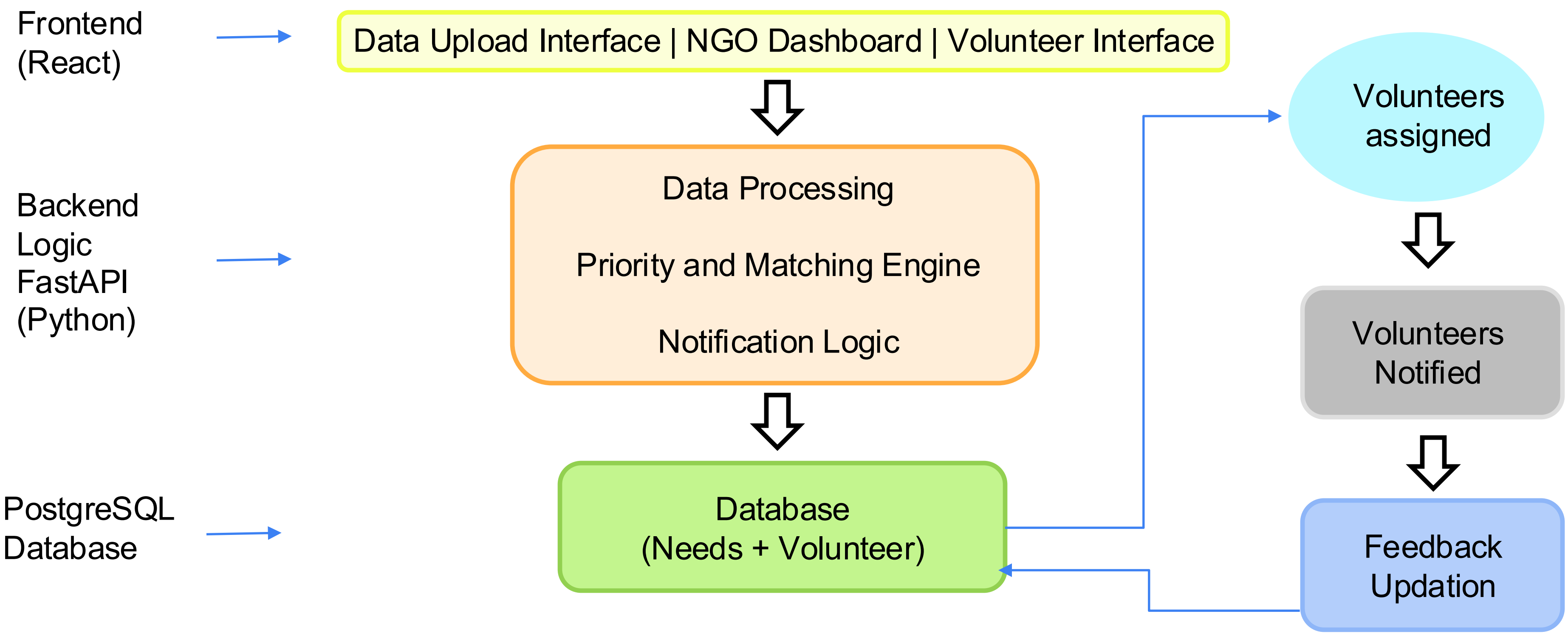
Before Matching



After Matching



Architecture diagram of the proposed solution



Technologies to be used in the solution

AI Processing

Gemini API – Interprets and prioritizes unstructured data

Backend

FastAPI (Python) - Handle APIs and core logic
Matching Engine – Rule-based scoring for optimal assignments

Database

PostgreSQL - Stores needs, volunteers and assignments

Frontend

React - User interface and interaction

Visualization

Leaflet Maps – Real-Time mapping of needs and volunteers

Notifications

Email (SMTP (Gmail)) - Alerts for assignments and updates

Estimated implementation cost (optional)

FEATURE	COST	REASON
1. Developmental Cost	₹ 0 (No external cost)	Built by the team
2. Hosting Frontend : Vercel Backend : Render Database: Render PostgreSQL	 ₹ 0 (Free Tier) ₹ 0 (Free Tier) ₹ 0 (Free Tier)	 Random domain name Free for low usage Free for a single database
3. Leaflet Maps	₹ 0 (Free Tier)	For demo purpose
4. Email SMTP (Gmail)	 ₹ 0 (Free Tier)	 500 emails/day

The system can be implemented at near-zero cost using free-tier cloud services, making it highly accessible for NGOs and small organizations.

However, as usage scales, additional investments may be required to support higher data volume.

Solution Challenge

Snapshots of the MVP

Quick Connect

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📍 Tambaram

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📍 Velachery

👥 Volunteers

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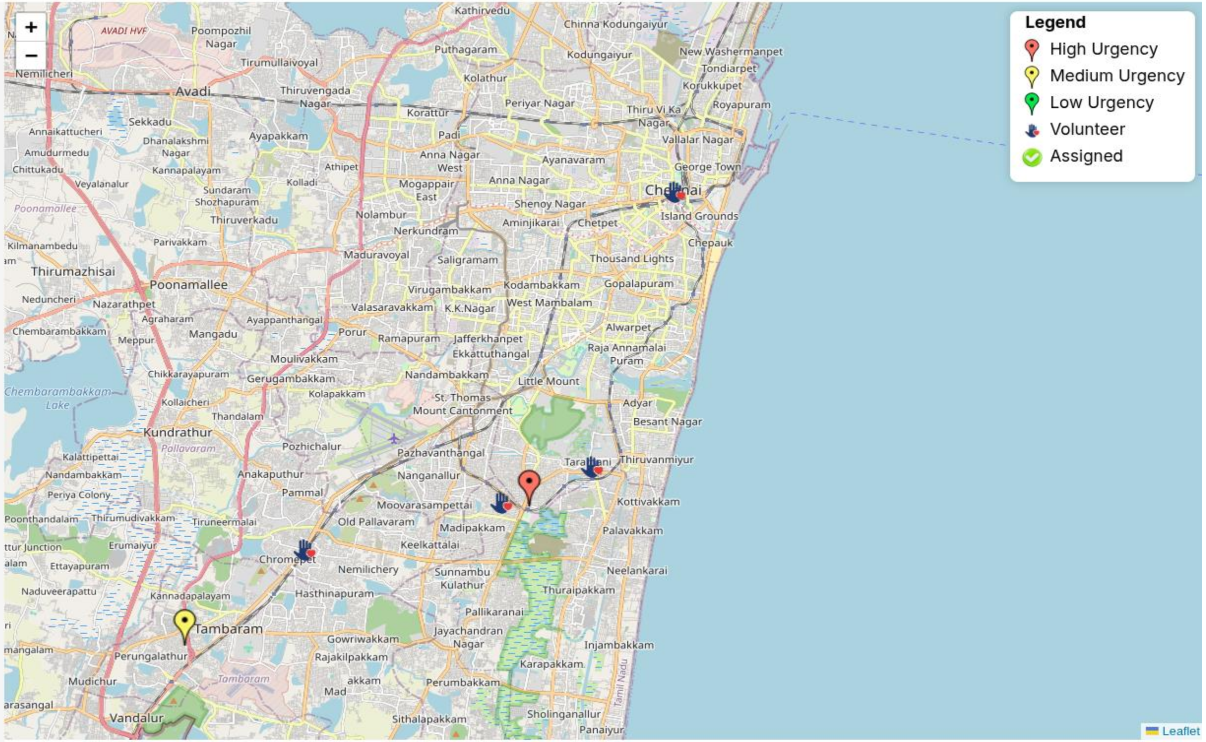
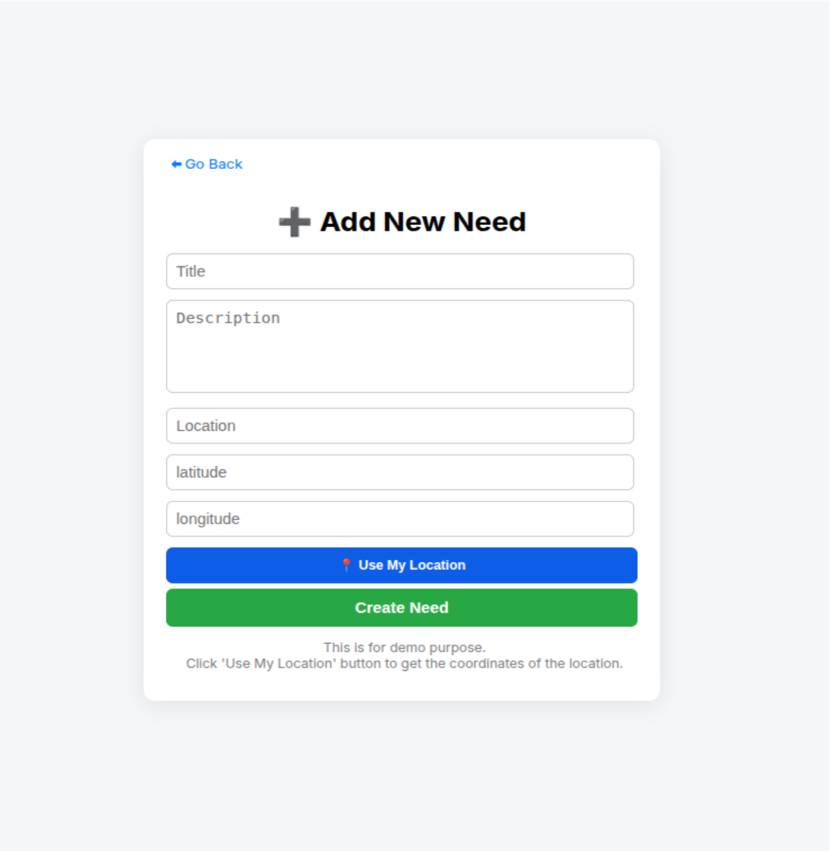
⚙️ Demo Purpose

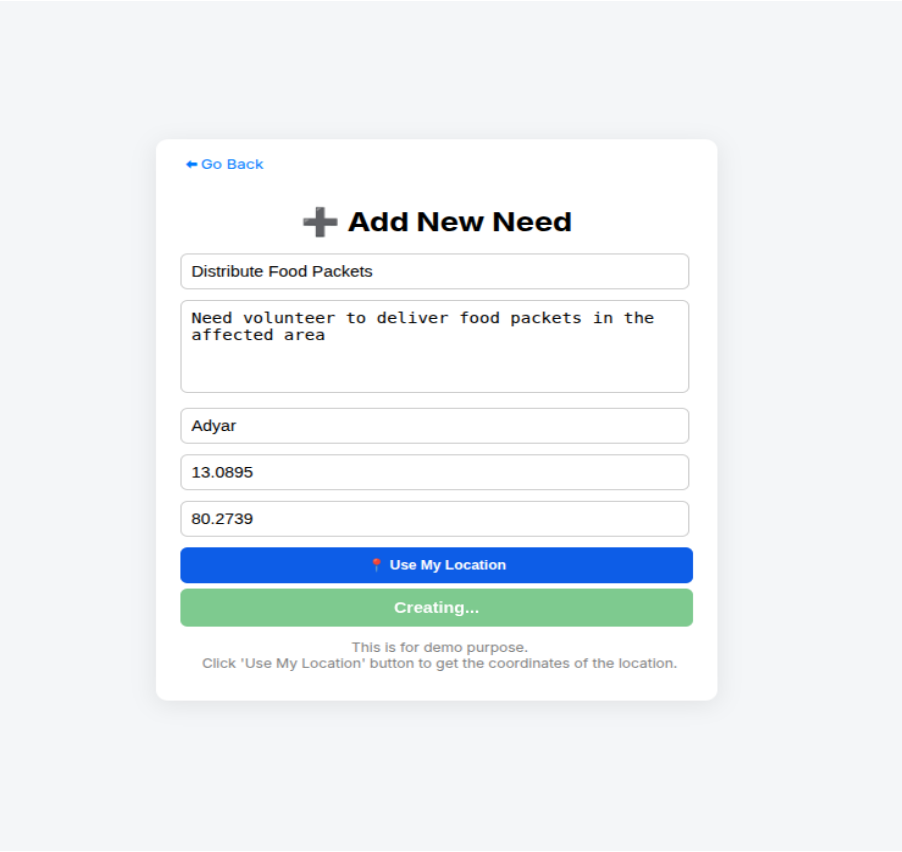
Clear All Assignments

➕ Add Need

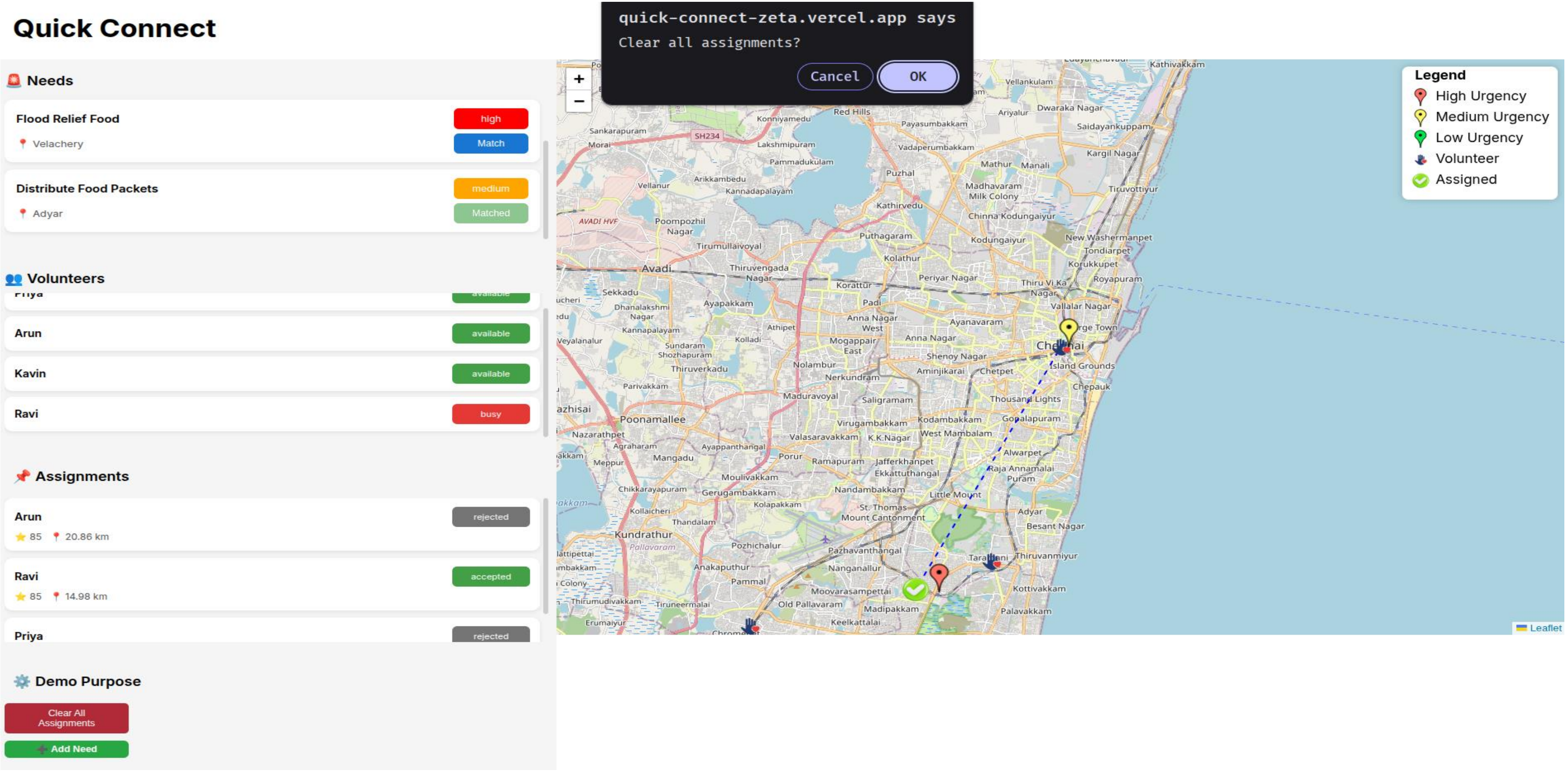


Snapshots of the MVP



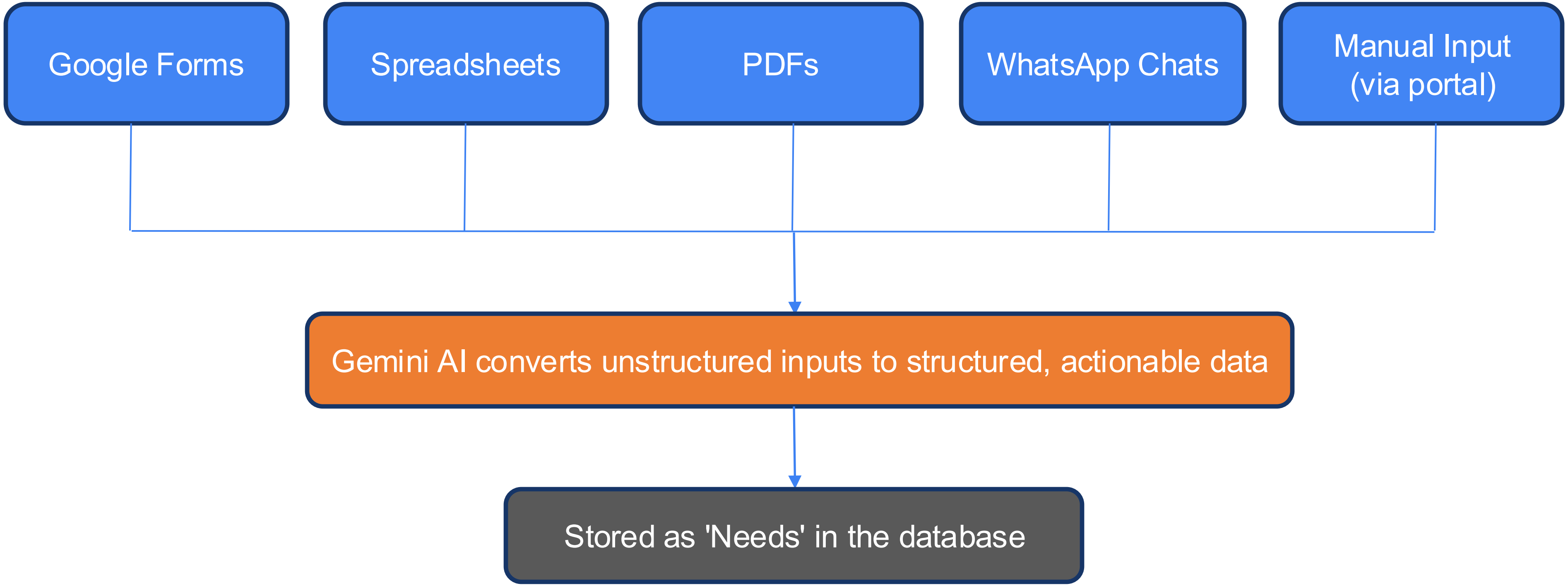


Snapshots of the MVP



Additional Details

1. Multi-Source Data Integration and Data Preprocessing



Additional Details

2. AI Interpretation and Prioritization

-> Converts unstructured inputs (texts, reports, messages) into Structured data

-> Identifies category (food, medical, shelter, rescue)

-> Determines urgency level from the given context

-> Outputs clean JSON for backend processing

Sample Input: "Need food urgently for 20 people"
Output:
{
"category": "food",
"urgency": "high"
}

AI Interpretation and Prioritization Logic

```
client = genai.Client(api_key=os.getenv("GEMINI_API_KEY"))

def classify_need(description: str):
    response = client.models.generate_content(
        model="gemini-flash-lite-latest",
        contents=f"""
        Return JSON only:

        category: food | medical | shelter | rescue
        urgency: low | medium | high

        Text: {description}
        """
    )

    text = response.text.strip().replace("```json", "").replace("```", "")

    try:
        parsed = json.loads(text)
        return parsed
    except:
        return {
            "category": "general",
            "urgency": "medium"
        }
```

Additional Details

3. Smart Scoring Engine:

Factors considered:

- Volunteer's Availability
- Proximity to the task
- Skill Relevance

Scoring Formula:

Score = 50 (Availability) + 20 (Distance) + 30 (Skill Relevance)

Weight Distribution:

- Availability -> 50%
- Distance -> 20%
- Skill Match -> 30%

Note:

Engine assigns **top 3 volunteers** for the task based on this score

Backend Scoring Logic

```python
if v.availability == "available":
    score += 50

if need.latitude and v.latitude:
    distance = calculate_distance(
        need.latitude, need.longitude,
        v.latitude, v.longitude
    )

    if distance < 5:
        score += 20
    elif distance < 10:
        score += 10
    else:
        score += 5

if need.category and v.skills:
    skills_list = [s.strip() for s in v.skills.lower().split(",")]
    if need.category.lower() in skills_list:
        score += 30
```

skills: Unknown

Additional Details

Note:

For demo purpose:

- **Web-based prototype** built to demonstrate system functionality
- **Unified dashboard** simulates end-to-end workflow
- **Manual input used** instead of multi-source ingestion (planned feature)
- **Location data captured** via browser (latitude & longitude)
- **Preloaded sample data** for Needs and Volunteers
- **Interactive controls** to simulate matching pipeline
- **Map visualization** displays static demo data from the given input

Future Development

1. **Expansion:**
   Mobile application for field workers
   Multi-NGO collaboration platform

2. **Intelligence:**
   Predictive Analysis for high-risk areas
   Advanced Matching (routing + workload balancing)

3. **Real-Time and Accessibility:**
   Live Volunteer tracking on map
   Whatsapp / SMS integration for data input

With these improvements, our system will evolve from a matching tool into a real-time crisis response platform.

URL Links:

1. GitHub Public Repository: https://github.com/Vaillant06/QuickConnect
2. Demo Video Link (3 Minutes): Demo_Video_Link
3. MVP Link: QuickConnect
4. Working Prototype Link: QuickConnect

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Thank You